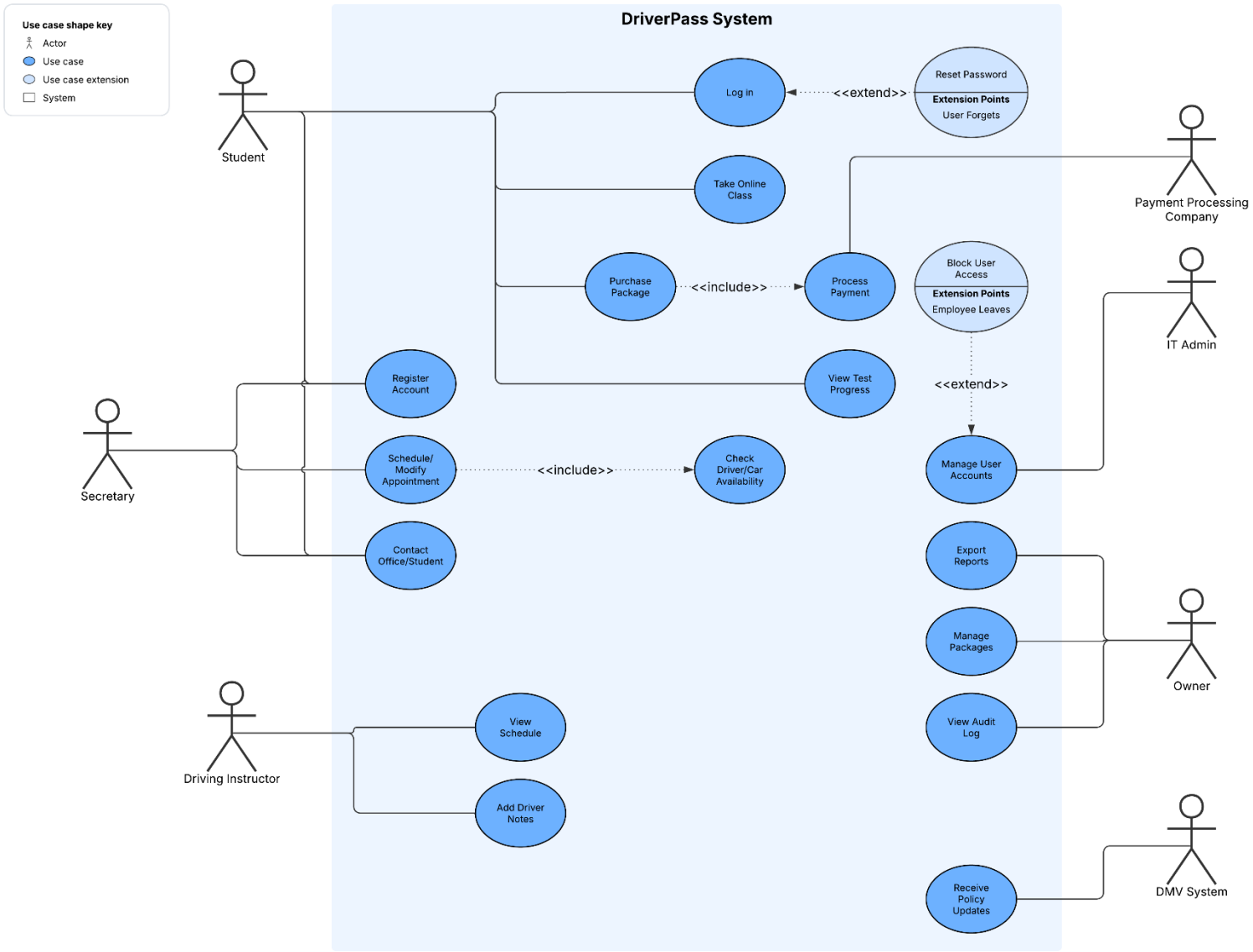
CS 255 System Design Document Template

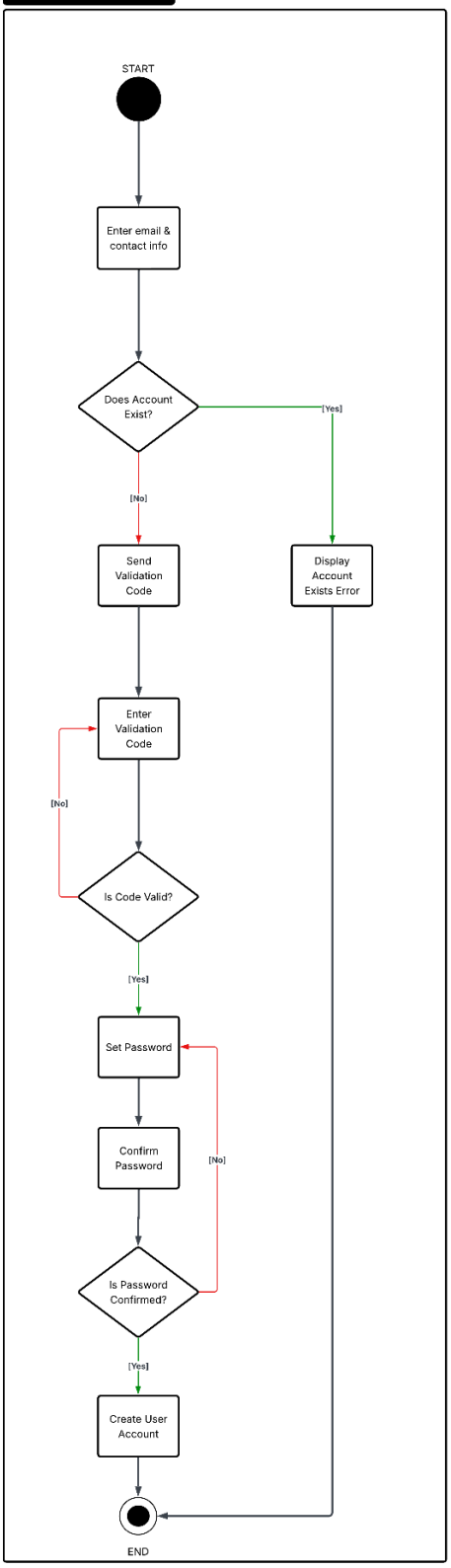
UML Diagrams

UML Use Case Diagram

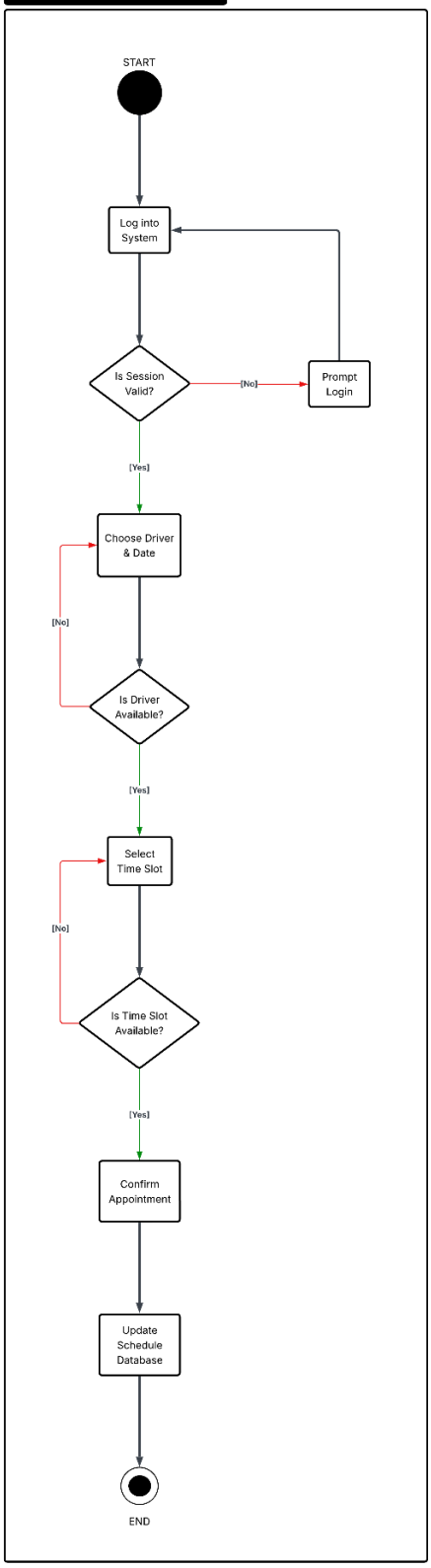


UML Activity Diagrams

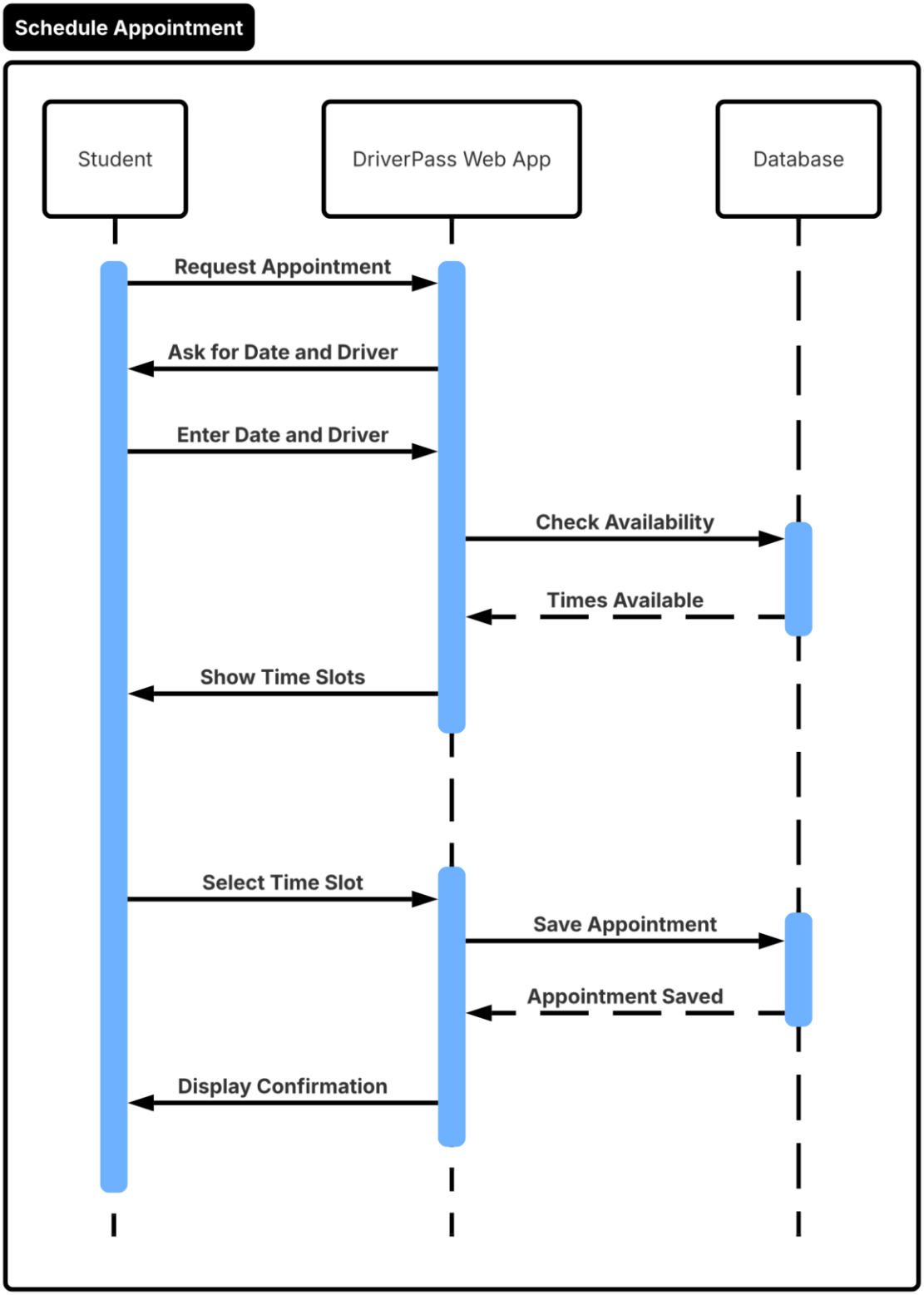
Register Account



Schedule Appointment

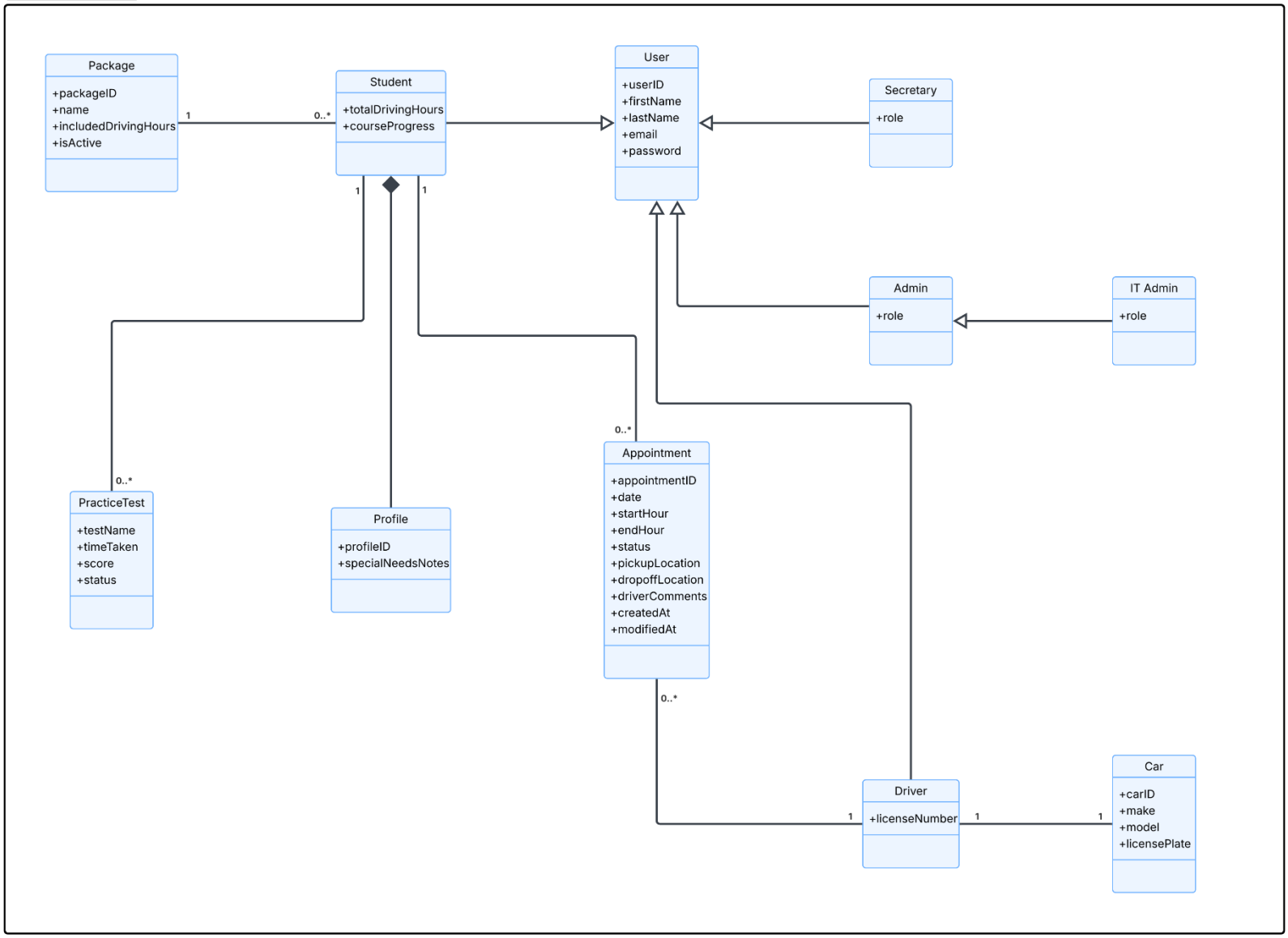


UML Sequence Diagram



UML Class Diagram

DriverPass UML Class Diagram



Technical Requirements

Infrastructure

The most important technical requirement for the DriverPass system is that it must be built entirely on a cloud-based infrastructure, as requested by the stakeholder. The cloud-based system will allow the owner, Liam, to access data from anywhere, and students will be able to log in on any modern device with access to the internet. Being on the cloud gives DriverPass the ability to scale up and out, meaning they can have more capacity to handle spikes in demand, and will be available online 24/7. This also means they don't need to worry much about purchasing and maintaining their own servers.

Hardware

As mentioned previously, since DriverPass will be a cloud-hosted, web-based platform, there isn't much hardware required to develop this system. The employees will need a computer and a phone to handle administrative tasks on their end, and the customers will, of course, need a computer or smartphone to access the internet.

Software

The frontend of the system will need to leverage JavaScript, HTML, and CSS to build the web portal interface that the students will be using. The backend of the system will require a modern, OOP-based programming language to program all the business logic required for account registration, user management, and scheduling, as shown in the diagrams above. We will also need a Relational Database System. As seen in the UML Class Diagram, the data is structured in such a way that the relationships, such as Student -> Appointment -> Driver -> Car, need to be tied together efficiently, and the data can be retrieved quickly for the front-end dashboards and the admin reports.

Tools

The system will need a few tools that are specialized for their individual functionality. We will need a 3rd-party payment gateway tool, like Stripe. This tool offloads all the complexities of handling sensitive financial data and communication with the banks from the DriverPass system, which is one of the requests that the owner made. We will, however, need a Data Encryption tool to encrypt and protect the data in transit during communication between the Customer and the System, as well as from the System to the Payment Processor. We'll also need to encrypt the data at rest in the database to protect Personally Identifiable Information (PII) that will be stored in the DriverPass Database. Finally, the last tool we'll need is an API tool to connect the DriverPass System directly to the DMV's databases so DriverPass can receive automated updates whenever policies change or new practice questions are released.